

OpenOrder Menu Architecture Audit — Integration-Neutral Domain

Date: 2026-07-24

Scope: Menu identity, import/publish, storefront browse, cart/modifiers, external mappings, order routing

Trigger: Square multi-variation items were hidden after publish because Square parent ITEM ids were stored in `deliverectVariantParentPlu` , a Deliverect-only browse/routing field.

Executive summary

OpenOrder already has a **shared canonical menu shape** (`MennyuCanonicalMenu`) and a **shared publish/storefront pipeline**. All three producers — Deliverect, Square, and the manual Open Order builder — emit that shape.

The architectural debt is that the shared shape and live DB columns use **Deliverect-first names** (`deliverectId` , `deliverectProductId` , `menu.deliverect` , `deliverectVariantParentPlu` , ...). Those names encode **provider-specific semantics** that generic layers then interpret incorrectly when another provider reuses them.

Finding	Severity
Stable product identity is named <code>deliverect*</code> but holds <code>sq:*</code> / <code>oo:*</code> / Deliverect ids	High (naming / ownership)
Browse visibility keys off <code>deliverectVariantParentPlu</code> (Deliverect leaf semantics)	High (proven production bug)
Square location written into <code>MenuImportJob.deliverectLocationId</code>	Medium (F-class reuse)
Square source meta nested under <code>menu.deliverect</code>	Medium (F-class reuse)
<code>ProviderEntityMapping</code> already approximates <code>ExternalMenuMapping</code> for Square	Positive
Deliverect still uses embedded <code>Menuitem</code> columns as mapping SoT	Expected today; hybrid long-term
Shared import UI still says “Deliverect” for Square/manual jobs	Medium (UX / ops confusion)

Immediate fix already shipped (pre-audit): Square no longer sets `deliverectVariantParentPlu` ; parent ITEM id lives on `sourceParentExternalId`

in the canonical snapshot. This audit defines the long-term domain and a phased migration so the class of bug cannot recur.

This pass implements only Phase 1 (safe): generic identity helpers, provider consistency diagnostics, source-aware UI copy, and regression tests. **No destructive schema migration; no legacy column deletes.**

Phase 1 implementation status (completed in this change set)

Deliverable	Location
Identity helpers	src/domain/menu-import/canonical-identity.ts
sourceProvider derivation	src/domain/menu-import/menu-source-provider.ts
Consistency diagnostics	src/domain/menu-import/menu-provider-consistency.ts (+ import review UI)
Neutral shared import copy	MenuImportPublishPanel , IssuesList, ParityBanner, AdvancedDetails, DiffView, labels
Tests	menu-provider-consistency.test.ts (+ existing Square visibility suite)
Interactive summary	Cursor canvas menu-architecture-audit.canvas.tsx

1. Inventory & classification legend

Code	Meaning
A	Deliverect-only integration concern
B	Square-only integration concern
C	Generic OpenOrder menu concern
D	Order-routing concern
E	Legacy naming (generic concept wearing a Deliverect name)
F	Incorrect cross-integration reuse

2. Field ownership matrix

2.1 Canonical snapshot (MennyuCanonicalMenu)

Field / concept	Class	Written by	Read by	Notes / risks
<code>products[].deliverectId</code>	E	All adapters	Publish, browse, diffs, storefront	Generic entity id; name is misleading
<code>categories[].deliverectId</code>	E	All adapters	Publish, storefront sections	Same
<code>modifierGroupDefinitions[].deliverectId</code>	E	All adapters	Publish, links	Same
<code>options[].deliverectId</code>	E	All adapters	Publish, Square PEM	Same
<code>productDeliverectIds</code>	E	All adapters	Browse, preview	Same
<code>menu.deliverect</code> meta object	E / F	All adapters	Source detection, Square readiness	Square/OC meta live under Deliverect-named bag
<code>sourcePayloadKind</code>	C	All adapters	Source routing	Real general discriminator wrong parameter object name
<code>deliverectVariantParentPlu / Name</code>	A	Deliverect only (Square must be null)	Browse exclusion, cart nesting, Deliverect transform	Must not used for Square parents
<code>sourceParentExternalId</code>	B→C	Square	Snapshot diagnostics / future mapping	Not on live <code>MenuItem</code> yet

plu / option plu	A	Deliverect	Deliverect orders, snooze	Square leaves null
isVariantGroup / multiMax	A (semantics)	Deliverect; optional elsewhere	Modifier kinds, Deliverect transform	Square typically unset

2.2 Live tables (MenuItem / ModifierGroup / ModifierOption)

Column	Class	Role
MenuItem.deliverectProductId	E	Upsert key + Square PEM internalEntityId
MenuItem.deliverectPlu	A	Deliverect PLU / snooze
MenuItem.deliverectVariantParentPlu / Name	A	Deliverect leaf → browse hide + cart/order nesting
MenuItem.deliverectCategoryId	E	Category link for live rows
ModifierGroup.deliverectModifierGroupId	E	Upsert identity
ModifierGroup.deliverectIsVariantGroup / MultiMax	A	Deliverect nesting rules
ModifierOption.deliverectModifierId	E	Upsert + Square PEM
ModifierOption.deliverectModifierPlu	A	Deliverect outbound PLU

2.3 Import / job / vendor

Field	Class	Notes
MenuImportJob.source	C	Discriminator (SQUARE_CATALOG_PULL , DELIVERECT_*, ...)
MenuImportJob.deliverectChannelLinkId	A	Deliverect-only
MenuImportJob.deliverectLocationId	F when Square	Square writes Square location id here today
MenuImportJob.deliverectMenuId	A	Deliverect-only
MenuImportIssue.deliverectId	E / F	Square writes squareObjectId into it

<code>Vendor.menuSource</code>	C (incomplete)	Enum is only <code>open_order</code> <code>deliverect</code> ; Square runs as <code>open_order</code> + <code>orderRoutingMode=square</code>
<code>Vendor.orderRoutingMode</code>	D	Separates routing provider from menu ownership (good)
<code>ProviderEntityMapping.*</code>	C	Square ExternalMenuMapping spine
<code>VendorIntegrationConnection.externalLocationId</code>	C	Correct generic location

2.4 Browse / storefront / cart

Concept	Class	Notes
<code>computeCustomerMenuBrowseExcludedProductIds</code>	C with A -shaped rule	Excludes products with variant-parent PLU set — correct for Deliverect leaves; catastrophic if Square reuses the field
Storefront section builder	C	Should never branch on provider
Cart variant resolution (<code>cart-deliverect-variant-resolution</code>)	A / D	Deliverect leaf merge; gated by Deliverect nesting rules for orders
Square order mapper	D	Reads <code>deliverectProductId</code> as Square internal key → PEM

3. Risky cross-provider usages (F-class and proven bugs)

#	Risk	Location	Status
1	Square parent ITEM id written as <code>deliverectVariantParentPlu</code> → storefront hide	Former Square normalizer	Fixed (null + <code>sourceParentExternalId</code>)
2	Square catalog ids stored in <code>deliverectId</code> / <code>deliverectProductId</code>	Square normalizer + publish	Ongoing E ; rename later
3	Square location → <code>MenuImportJob.deliverectLocationId</code>	<code>square-menu-import.service.ts</code>	F ; Phase 2 column/alias
4	Square warnings → <code>MenuImportIssue.deliverectId</code>	Same	F ; Phase 2

5	Square meta under <code>menu.deliverect</code>	Square normalizer	F ; Phase 2 rename meta bag
6	Shared UI: “Fix in Deliverect” / “Removed items in Deliverect”	Menu import components	Phase 1 copy cleanup
7	Helpers named <code>isSquareProductDeliverectId</code>	<code>square-menu-ids.ts</code>	Phase 1 alias rename in TS

Invariant (must remain tested forever):

A field used for **order nesting / Deliverect leaf semantics** must not be overloaded as **generic parent external id**. Browse visibility must only depend on an explicitly modeled “variant leaf / non-browsable” flag — today that flag is `deliverectVariantParentPlu`, which is Deliverect-shaped.

4. Generic OpenOrder menu domain (target)

4.1 Core concepts

Concept	Responsibility
Menu (<code>MenuVersion</code> + canonical snapshot)	Versioned, vendor-owned catalog snapshot; draft vs published
MenuCategory	Browse grouping; ordered list of item identities
MenuItem	Customer-sellable SKU (price, availability, modifiers, image)
MenuItemVariation	<i>Not</i> a separate live table today. Square flattens variations into Menuitems. Deliverect models variations as leaf Menuitems under a parent shell. Retain as an optional conceptual distinction in adapters, not a required core table in Phase 1–2.
ModifierGroup / ModifierOption	Selection rules and priced choices
MenuItemModifierGroup	Link table: which groups apply to which item
ExternalMenuMapping	Provider ↔ internal entity mapping (see §6)
MenuImport / MenuImportDraft	Ingest job + draft <code>MenuVersion</code>
MenuPublish	Draft → live tables + published pointer
Browse / orderability state	Visibility on storefront vs cart/order eligibility (availability, snooze, mapping readiness)

4.2 Core vs adapter-owned fields

Core domain (generic):

name, description, priceCents, isAvailable, sortOrder, imageUrl, basketMaxQuantity, category membership, modifier group links, min/max/required selection rules, published/draft state, vendor ownership.

Adapter-owned (must not leak into generic browse/cart rules unless explicitly modeled):

Deliverect PLU, channel link, variant parent PLU/name, `isVariantGroup` / `multiMax` , Square ITEM vs ITEM_VARIATION ids, Square location/merchant, OAuth connection ids, raw provider payloads.

Explicitly modeled cross-cutting flags (generic):

`isVariantLeaf` / non-browsable (today implied by non-null variant parent PLU) — should become a **named generic boolean or enum** in Phase 2 so Deliverect can set it and Square never touches Deliverect PLU fields.

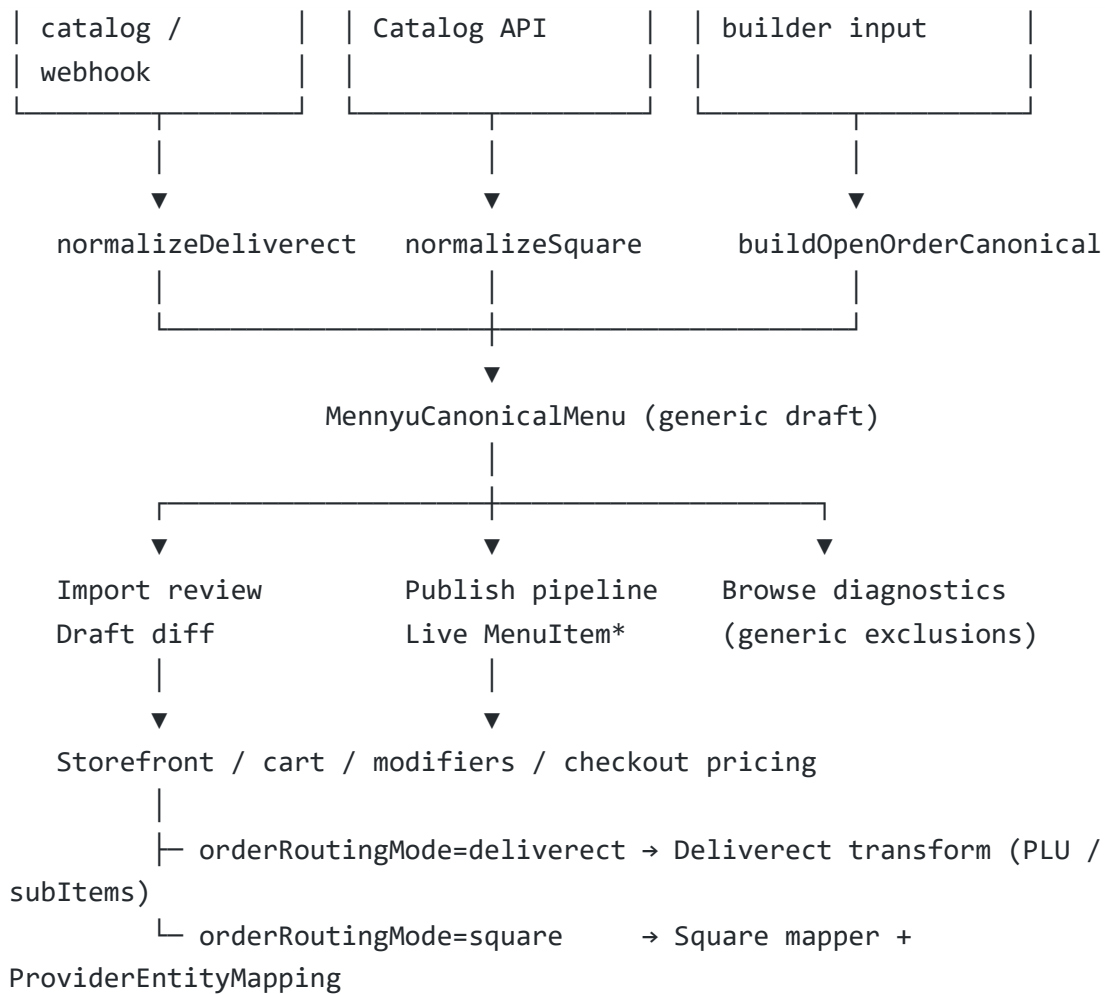
5. Generic external-source terminology (target)

Prefer	Avoid for generic / Square / manual
<code>sourceProvider</code>	Hardcoding "Deliverect" in shared UI
<code>sourceExternalId</code>	<code>deliverectId</code> as the public name
<code>sourceParentExternalId</code>	Storing parents in <code>deliverectVariantParentPlu</code>
<code>sourceLocationId</code>	<code>MenuImportJob.deliverectLocationId</code> for Square
<code>sourceAccountId</code>	Mixing merchant ids into menu columns
<code>sourceVersion</code> / <code>sourceMetadata</code>	Overloaded PLU columns
<code>externalMapping</code> / <code>ProviderEntityMapping</code>	Embedding all provider ids forever on MenuItem

JSON snapshot keys and Prisma columns keep legacy names through Phase 3; TypeScript helpers and UI use the preferred terms from Phase 1.

6. Adapter boundaries





Rule: Generic layers consume only generic menu concepts. Provider branching is allowed only at:

1. Adapter normalizers (ingress)
2. Order-routing adapters (egress)
3. Provider-specific settings / diagnostics pages

7. External mapping architecture recommendation

Option A — Embedded fields only (status quo for Deliverect)

Pros: simple joins; publish upsert by column.

Cons: every new POS adds columns or overloads `deliverect*` ; proven semantic collisions.

Option B — Mapping table only

Pros: clean provider isolation.

Cons: large migration; every storefront/cart path must join mappings; Deliverect production risk.

Option C — Hybrid (recommended)

Concern	Home
Stable internal upsert key on live rows	Keep thin embedded key (rename later to <code>sourceEntityId</code> / keep prefixed values)
Provider catalog object id + location + parent	<code>ProviderEntityMapping</code> (already exists) / evolve toward recommended shape
Deliverect PLU / leaf / variant-group	Stay Deliverect-specific (embedded or Deliverect side-car) — never reused by Square
Parent Square ITEM id	Mapping <code>externalParentId</code> or metadata; snapshot <code>sourceParentExternalId</code> until then

Recommended target mapping shape (evolve `ProviderEntityMapping` , do not invent a parallel table in Phase 1):

- `vendorId` , `provider` , `environment`
- `externalAccountId` , `externalLocationId`
- `entityType` , `internalEntityId` , `externalId` , `externalParentId`
- `externalVersion` , `isActive` , `metadataJson`

Safest long-term path: Hybrid. Square already uses PEM for injection. Deliverect keeps embedded SoT until a deliberate dual-write. Do **not** auto-create a second mapping table now.

8. Routing separation

Concern	Must not auto-control
Menu identity (<code>deliverectProductId</code> / future <code>sourceEntityId</code>)	Browse visibility
External mapping (PEM)	Browse visibility
Order-routing payload construction	Storefront tile list
<code>deliverectVariantParentPlu</code> (Deliverect leaf)	Square parent ITEM linkage

Explicit model: “non-browsable / variant leaf” is a **menu presentation** concern. “parent external id for POS” is a **mapping** concern. They coincide for Deliverect leaves today; they must not for Square.

9. Naming cleanup (UI / code)

Current	Replacement
"Fix these in Deliverect..."	"Fix these in the source menu system..." / source-aware ("Square catalog" / "Deliverect" / "Open Order")
"Removed items in Deliverect..."	"Removed items from the imported menu are marked unavailable, not deleted."
"identical Deliverect ids"	"identical external catalog ids"
"Deliverect mapping" (parity banner, shared)	"External menu mapping" / "Published menu mapping"
<code>hideDeliverectIds</code>	<code>hideExternalIds</code> (alias)
Advanced "Deliverect" for location when Square	"Source location" / "Square location"
<code>productDeliverectId</code> in diagnostics	Keep JSON key; display "Source item ID"

Keep "Deliverect" labels only on Deliverect connect, channel, pull, and Deliverect-only diagnostics.

10. Phased migration plan

Phase 1 — Introduce generics; preserve behavior (*this pass*)

- Generic identity / provider helpers (TS)
- Consistency diagnostics (no secrets)
- Source-aware / neutral UI copy on shared import surfaces
- Regression tests (Square visibility, Deliverect leaves, manual menus, browse vs routing separation)
- **No DB migration; no column deletes; snapshot JSON keys unchanged**

Rollback: Revert code; no data migration to undo.

Phase 2 — Writers + readers + backfill

- Dual-write optional generic fields in snapshot (`sourceProvider` , aliases) while keeping legacy keys
- Square/Deliverect/manual writers emit both
- Generic readers prefer new helpers
- Backfill diagnostics for mismatches
- Consider additive Prisma columns (`sourceEntityId` nullable) **only with dual-write** — separate approval

Rollback: Stop writing new fields; readers fall back to legacy keys. Additive columns remain nullable unused.

Phase 3 — Remove generic dependencies on legacy Deliverect names

- Generic browse uses `isVariantLeaf` (or equivalent) rather than Deliverect PLU field name in code paths
- Deprecate reading `menu.deliverect` without helper
- Mark Prisma columns deprecated in schema comments

Rollback: Helpers still read legacy columns.

Phase 4 — Delete / rename legacy fields

- Only after zero reads/writes proven (grep + telemetry + diagnostics clean)
- Rename/migrate DB columns with expand/contract
- **Requires explicit approval**

Rollback: Restore previous schema from backup; keep expand phase columns until contract proven.

11. Consistency checks (diagnostics)

Safe checks (no tokens / credentials):

1. Square-sourced products with non-null `deliverectVariantParentPlu`
2. Deliverect-sourced leaves missing parent PLU when snapshot expects leaves
3. `sourcePayloadKind` vs job `source` mismatch
4. Active PEM rows for non-selected Square location
5. Duplicate active PEM rows for same internal entity + location
6. Published items excluded from browse solely due to variant-parent metadata
7. Modifier groups linked only to unavailable/stale items

Implemented in Phase 1 as pure functions over canonical menus (+ optional PEM inputs later).

12. Exact files to change

Phase 1 (safe — implemented / in this pass)

Area	Files
Domain helpers	<code>src/domain/menu-import/canonical-identity.ts</code> (new), <code>menu-source-provider.ts</code> (new), <code>menu-provider-consistency.ts</code> (new)

Browse copy	<code>src/domain/menu-import/customer-menu-browse.ts</code>
Shared UI	<code>MenuImportPublishPanel.tsx</code> , <code>MenuImportIssuesList.tsx</code> , <code>MenuParityAuditBanner.tsx</code> , <code>MenuImportMenuPreview.tsx</code> , <code>MenuImportAdvancedDetails.tsx</code> , <code>MenuImportBrowseExclusionDiagnostics.tsx</code>
Labels	<code>vendor-menu-import-labels.ts</code> , admin/vendor import pages (copy)
Tests	<code>canonical-identity.test.ts</code> , <code>menu-provider-consistency.test.ts</code> , extend Square/Deliverect browse tests

Phase 2+ (approval required for schema)

Area	Files
Canonical schema dual keys	<code>canonical.schema.ts</code>
Normalizers	<code>square-catalog-normalizer.ts</code> , <code>integrations/deliverect/menu/normalize.ts</code> , <code>open-order-menu-publish.service.ts</code>
Publish	<code>menu-publish-from-canonical.service.ts</code>
Jobs	<code>MenuImportJob</code> Prisma columns / Square import writer
Cart	Gradually route through <code>isVariantLeafProduct()</code>
PEM evolution	<code>provider-mapping.service.ts</code> , <code>schema.prisma</code>

13. Tests to add / keep

Test	Proves
Square multi-variation visibility	Two variations visible; category remains; modifiers survive; PEM ids retained
Deliverect variant leaves	Still excluded from browse when parent PLU set
Manual / OO builder	No required external mapping fields for browse
Consistency diagnostic	Flags Square+parentPlu misuse
Generic publish identity helpers	Same external id accessors across providers
Browse vs routing	Parent external id $\neq$ browse exclusion

Stale PEM	Covered by existing Square mapping/location tests; keep
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14. Low-risk cleanup (Phase 1) vs high-risk (needs approval)

Implement now (Phase 1)

- TS helpers: `productExternalId` , `isVariantLeafProduct` , `menuSourceProvider` , `menuSourceMeta`
- Neutral / source-aware UI strings on shared import panels
- Diagnostics for Square entities using Deliverect-only leaf fields
- Regression tests listed above
- Alias `hideExternalIds` → existing `hideDeliverectIds`

Requires separate approval

- Prisma renames / new required columns
 - Changing published snapshot JSON keys without dual-write
 - Deleting `deliverect*` columns
 - Moving Deliverect SoT fully into PEM
 - Changing `Vendor.menuSource` enum (add `square`) without full call-site audit
 - Making `MenuItemVariation` a first-class table
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15. Recommended target schema (conceptual)

```
MenuVersion { canonicalSnapshot /* generic */, state, ... }

MenuItem {
  id, vendorId, name, priceCents, isAvailable, ...
  sourceEntityId           // Phase 2+ rename of deliverectProductId (dual-
                             write)
  isVariantLeaf            // Phase 2+ explicit; today derived from variant
                             parent PLU
  // Deliverect-only (stay namespaced or side-car):
  deliverectPlu, deliverectVariantParentPlu, deliverectVariantParentName
}

ProviderEntityMapping / ExternalMenuMapping {
  vendorId, provider, environment,
  externalAccountId, externalLocationId,
```

```
entityType, internalEntityId, externalId, externalParentId?,
externalVersion?, isActive, metadataJson?
}
```

16. Decision record

Decision	Choice
Long-term mapping	Hybrid — PEM for injection providers; thin embedded upsert keys; Deliverect-specific fields stay namespaced
Phase 1 schema	No migration
Browse control	Must not use provider parent-id fields; only explicit leaf/non-browsable semantics
Square parent ITEM	sourceParentExternalId (snapshot) → later PEM externalParentId

Appendix A — Provider boundary diagram (text)

See §6.

Appendix B — Related incidents

- 2026-07-24: Square multi-variation storefront disappearance via `deliverectVariantParentPlu` overload.
- Prior: Square location / PEM drift (Poke Sea) — mapping table location scoping is correct; reinforces PEM as routing SoT.